

PIONEER JUNIOR COLLEGE
JC2 Preliminary Examinations
In preparation for General Certificate of Education Advanced Level
Higher 2

CANDIDATE
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BIOLOGY

9744/03

Paper 3 Long Structured and Free-response Questions

18 Sep 2017

2 hours

Additional Materials: Writing Paper

READ THESE INSTRUCTIONS FIRST

Write your name, CT class and index number on all the work you hand in.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graph or rough working.
Do not use staples, paper clips, glue or correction fluid.

The use of an approved scientific calculator is expected, where appropriate. All workings and appropriate units must be shown.

The number of marks is given in brackets [] at the end of each question or part question.

This document consists of 2 sections:

Section A:

Answer **ALL** questions.

Answers are to be written in the spaces provided.

Section B:

Answer **ONE** question.

Write your answers on the separate writing papers provided.

Please hand in section A and section B separately.

Do not open this booklet until you are told to do so.

For Examiner's Use	
Section A	
1	/ 28
2	/ 10
3	/ 12
Section B	
4 or 5	/ 25
Total	/ 75

Section A

Answer **all** questions in this section.

- 1** Mosquito vectors *Aedes aegypti* and *Aedes albopictus* are main vectors of dengue virus (DENV) and chikungunya virus worldwide.

With about 50-100 million reported cases annually, including 500 000 severe cases of dengue haemorrhagic fever (DHF) or dengue shock syndrome (DSS), DENV is the most prevalent mosquito-borne human virus worldwide.

- (a)** Outline the life cycle of *Aedes aegypti*. [2]

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Fig. 1.1 shows a graph on development time of immature mosquitoes to the adult reproductive stage. The data is based on studies in a laboratory with mosquitoes taken from a tropical forest.

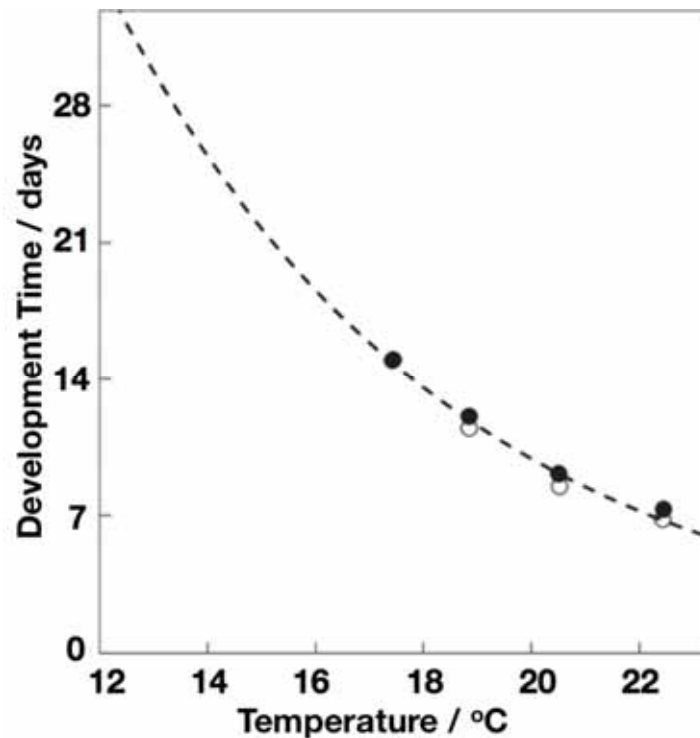


Fig. 1.1

(b) With reference to Fig. 1.1,

(i) explain how temperature changes impacts insects' metabolism. [2]

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(ii) explain the consequence of the trend on the spread of dengue virus. [2]

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The immune system is the body's defense against infectious organisms and other invaders. Through a series of steps called the immune response, the immune system attacks organisms and substances that invade body systems and cause disease.

(c) **(i)** Describe how the innate immune system normally responds to a microbial infection in the skin tissue, such as DENV. [3]

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- (ii) Explain why the normal innate immune responses prove to be ineffective when the body is infected with DENV. [2]

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- (d) In response to the DENV infection, the body's immune response reacts by producing antibodies that target the DENV virus.

- (i) Explain how the structure of the antibody allows for the successful recognition and binding of DENV virus in blood. [2]

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- (ii) Describe two ways in which the production of antibodies help in removing DENV from the body in the primary antibody response. [2]

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- (e) Despite the protection offered by the antibodies in the primary infection, the recurrent exposure to DENV, particularly of a different serotype, can result in the manifestation of severe dengue fever.

(i) State how the four different DENV serotypes differ. [1]

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(ii) Explain why the infection by a different serotype can result in severe dengue fever. [2]

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In 2016, the Health Sciences Authority (HSA) has approved the world's first dengue vaccine Dengvaxia for use in Singapore. Dengvaxia is a live, attenuated tetravalent dengue vaccine, and has shown to be effective in causing protection against the four DENV serotypes.

(f) Discuss two advantages of vaccination in the eradication of diseases such as dengue. [2]

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Besides vaccination, vector control programmes are widely adopted as a preventive measure. Unfortunately, these programmes are facing operational challenges with mosquitoes becoming resistant to commonly used insecticides in several areas through the world.

Spraying insecticide in regions with multiple stagnant water bodies is the main method of controlling *Aedes aegypti* in rural India. One of such insecticides was deltamethrin, which was introduced to rural areas in 2007.

A laboratory study was carried out using mosquitoes collected from two sites – A and B – in India. The percentage of mosquitoes killed by deltamethrin was estimated.

The results of the study are shown in Table 1.2.

Table 1.2

Site	Year	Percentage of mosquitoes killed by deltamethrin
A	2007	100
	2010	90
B	2007	100
	2010	100

- (g) The researchers concluded that at Site **A**, the mosquitoes had evolved resistance to deltamethrin. Explain how the mosquitoes evolved resistance. [3]

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India is one of the countries that has already been experiencing extreme weather events – extreme heat, droughts – due to climate change. Considering that agriculture play a vital role in India's economy, the impact of climate change on agricultural productivity has been a major concern.

Fig. 1.3 illustrates a prediction on global warming impacts on rice crop yield across India.

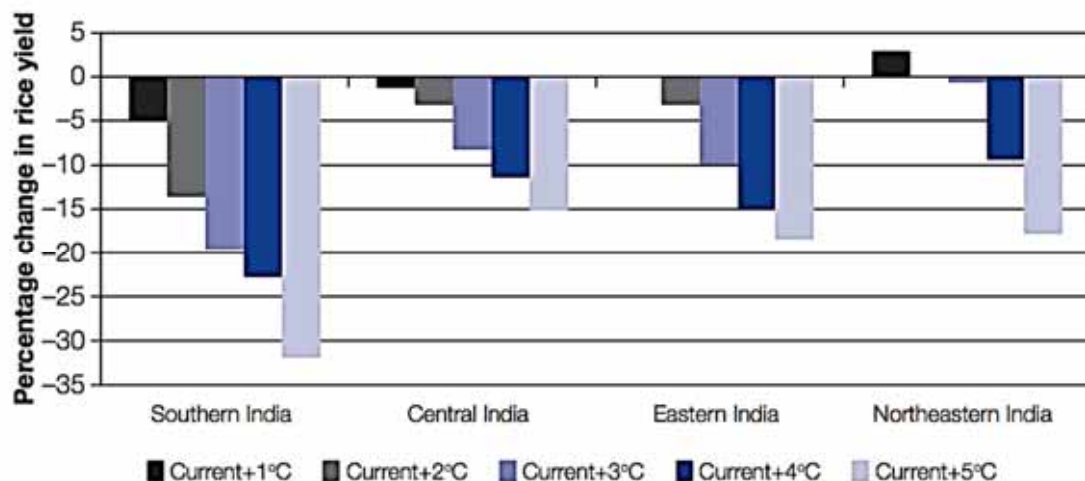


Fig. 1.3

In general, the trend is the similar for most plant crops.

(h) Explain the effects of increased temperature from climate change on plant crops. [2]

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It has been widely recognised that the effects of climate change have been brought about by excessive emission of greenhouse gases (GHG).

(i) A student made the following comment: 'If we stop deforestations, the concentration of GHG will decrease back to an acceptable level in the atmosphere'. Discuss the validity of this statement. [3]

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[Total: 28]

- 2 Fig. 2.1 shows the two distinct regions of human skin. The dermis is a thick region of living tissue below the epidermis, containing blood capillaries, nerve endings, sweat glands, hair follicles, and other structures.

The epidermis is composed of many different layers of different types of skin cells. These different layers of cells arose from the continual division and morphological changes of epidermal stem cells that are found in the basal layer of the epidermis.

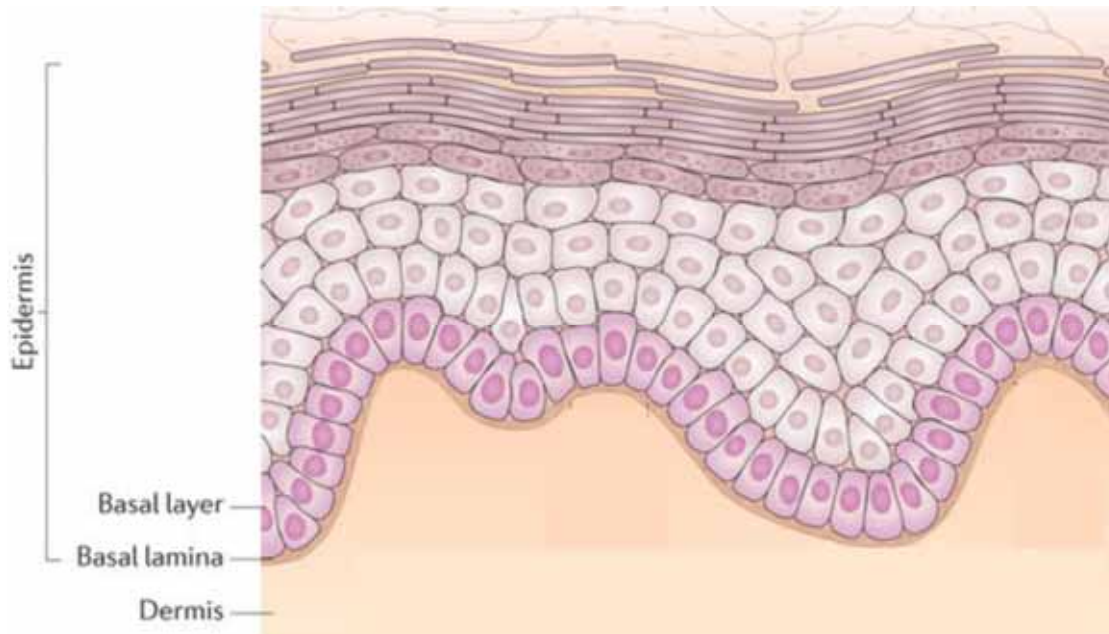


Fig. 2.1

- (a) With reference to Fig. 2.1,
 (i) describe the unique features of epidermal stem cells. [2]

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- (ii) state the potency of these epidermal stem cells. [1]

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(iii) explain the importance of the epidermal stem cells in the skin. [2]

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Until recently, burns have usually been treated with skin grafts, which involve taking skin sections from uninjured parts of the patient's body, and grafting them over the burn. These grafts can take several weeks or even months to heal, and during the recovery period, patients are prone to infections because of the damage to the skin, which is the body's first line of defence against microbes.

Scientists have now developed a new technique which involves harvesting the burn patient's skin stem cells, and stimulating them to divide using chemicals. These cells are then sprayed onto the burn. This method helps to regenerate the skin quickly, and dramatically reduce recovery times. Fig. 2.2 shows an illustration of this process.

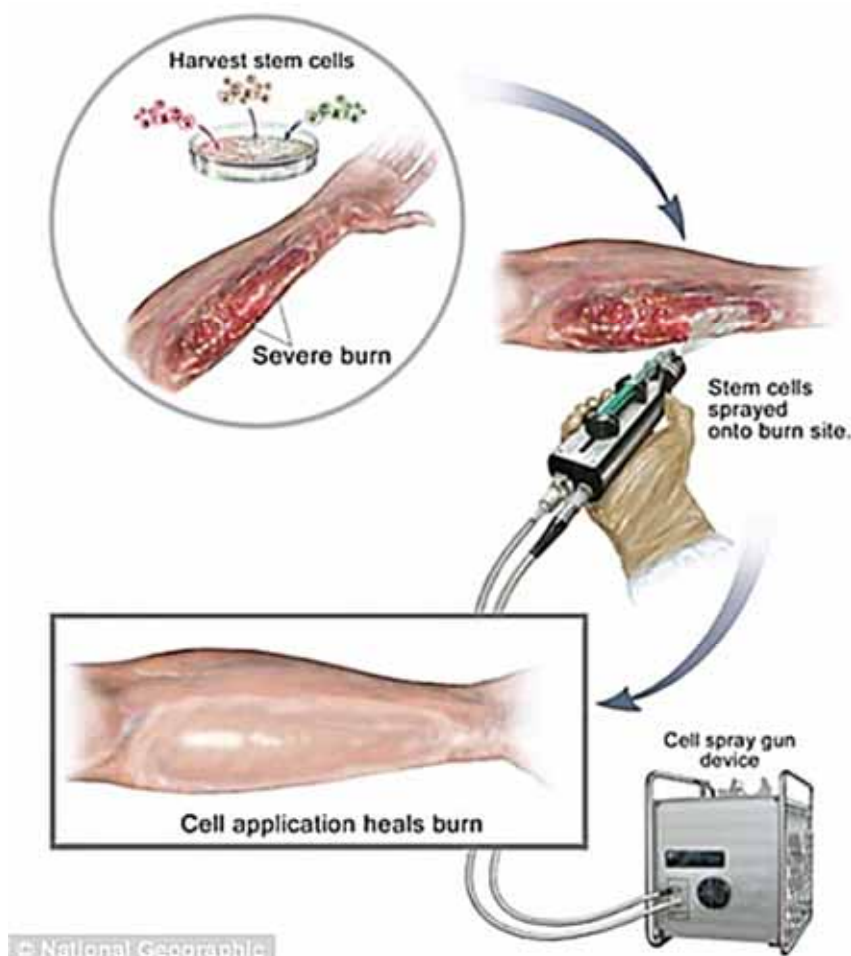


Fig. 2.2

Fig. 2.3 shows a photomicrograph of the skin stem cells undergoing repeated cell division in culture.

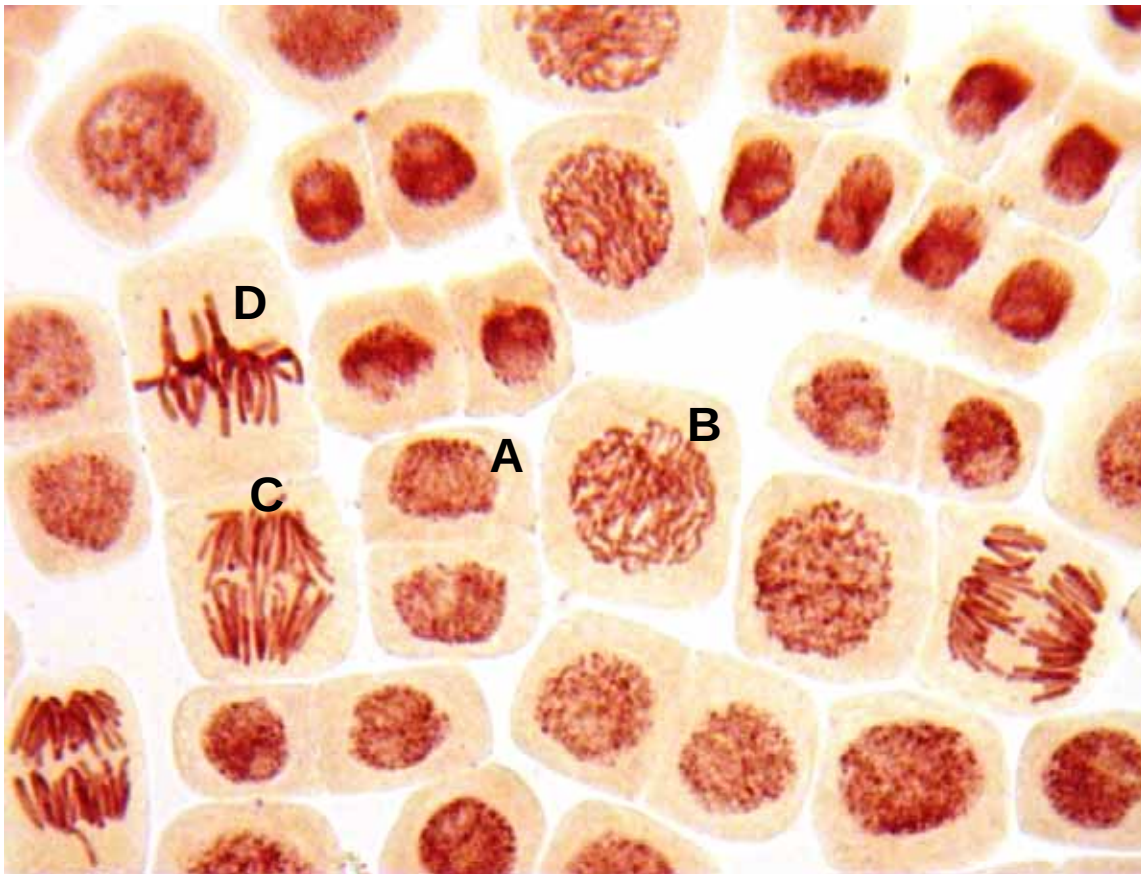


Fig. 2.3

(b) With reference to Fig. 2.3,

(i) arrange these stages in the correct sequence. [1]

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(ii) explain what is happening at stage **C**. [2]

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- (c) Suggest and explain why these stem cells need to be treated with chemicals to stimulate proliferation. [2]

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[Total: 10]

- 3 Fig. 3.1 shows a Siberian husky. The natural habitat of a Siberian husky is a cold, northern climate such as the Siberian Tundra or the wilds of Alaska. Siberian huskies were originally bred by the Chukchi people of Siberia to ultimately pull sleds across miles and miles of frozen ground. Basically, they were bred to be working dogs, as well as herd animals and perform as watchdogs.



Fig. 3.1

Fig. 3.2 represents the various birth weights of new-born puppies in a wild population of Siberian husky in Siberian Tundra. The line diagram on Fig. 3.2 represents mortality in relation to birth weight.

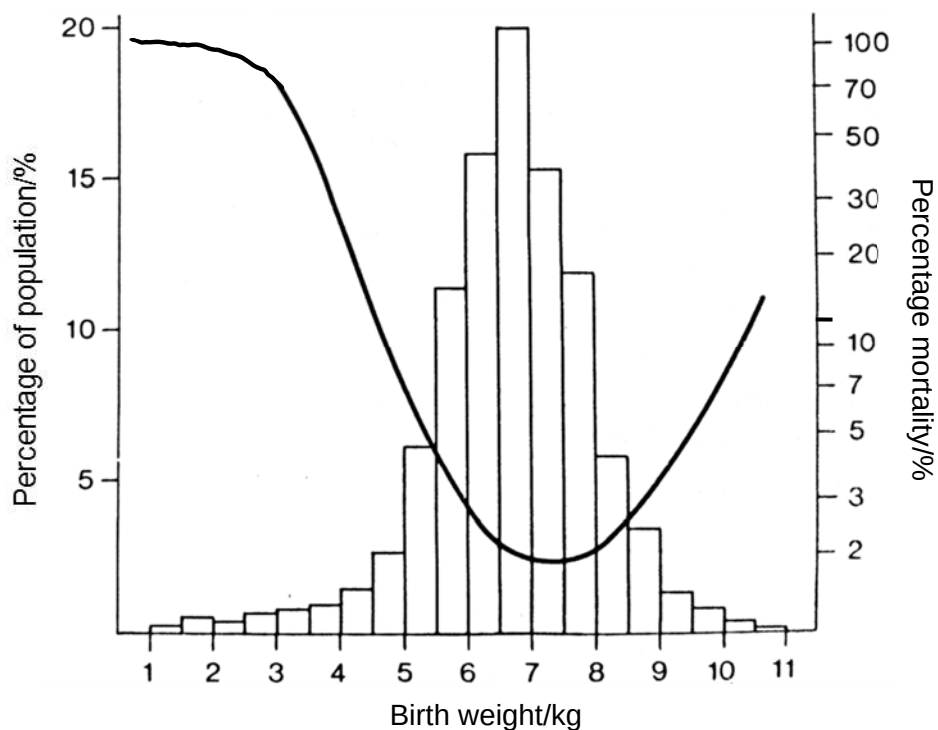


Fig. 3.2

- (a)** Using the information provided in Fig. 3.2, account for the type of selection acting on birth weight of new-born Siberian husky puppies. [3]

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- (b)** Birth weight of new-born Siberian husky puppies is an example of continuous variation. Explain why there is a variation of birth weights in the population of Siberian husky. [3]

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- (c)** Suggest why percentage of mortality is higher on both ends of the range of birth weights of new-born Siberian husky puppies. [2]

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(d) Suggest why Siberian husky and coyote are classified as different species. [2]

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(e) Some scientist studied the anatomical structures of the golden jackal and hypothesized that the golden jackal is more closely related to the Siberian husky than the grey wolf. Fig. 3.3 shows a segment of homologous DNA sequences from the golden jackal, coyote, grey wolf and Siberian husky.

	1	2	3	4	5	6	7	8	9	10	11	12	13	14
golden jackal	A	G	C	T	G	T	C	G	A	T	T	C	C	A
coyote	A	G	C	T	A	T	G	G	A	A	T	C	G	A
grey wolf	T	G	C	T	A	T	G	G	A	T	T	C	C	T
Siberian husky	T	G	G	T	A	T	G	G	A	T	T	C	C	A

Fig. 3.3

Suggest if the hypothesis that the golden jackal is more closely related to Siberian husky than the grey wolf is true. [2]

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[Total: 12]

Section B

Answer **ONE** question in this section.

Write your answers on the separate writing paper provided.

Your answers should be illustrated by large, clear labeled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

- 4 (a)** With reference to named examples, describe the range of roles performed by the proteins in living organisms. [13]
- (b)** Describe how Southern blotting can be used to analyse nucleic acids. [12]
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- 5 (a)** With reference to named examples, describe the range of roles performed by ATP in living organisms. [13]
- (b)** Compare and contrast between oxidative phosphorylation and photophosphorylation. [12]

END OF PAPER

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